Marta Kryven, Ph.D.

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Research Employment

2024 – **Assistant Professor**, Assistant Professor of Computer Science, cross-appointment with Psychology & Neuroscience, Dalhousie Dalhousie University, Canada
Director of Computation and Cognition Lab
Research interests: computational cognitive science; approximate Bayesian inference; AI interpretability, alignment and social cognition; modeling cognition in planning and long-horizon reasoning;
Advised 6 graduate and 7 undergraduate students

2018 – 2023 **Postdoctoral Research Scientist**, MIT, Brain and Cognitive Sciences, USA
Advisor: Josh Tenenbaum
Research interests: computational cognitive science, representation learning, planning, cognitive maps, computational perception
Advised 5 graduate students, and 11 undergraduates

Education

2012 – 2017 **Ph.D., University of Waterloo** Computer Science.
Thesis title: *Attributed Intelligence.*
Research interests: computational models of social perception, planning, and log-horizon reasoning; psychophysics, eyetracking, statistical modelling; experiment design;

2017 **Visiting Scholar, Massachusetts Institute of Technology**
Advisor: Prof. Joshua Tenenbaum.

2015 **'Brains, Minds Machines', Summer School**, The Marine Biological Laboratory, Woods Hole

Teaching Fundamentals Certificate, Centre for Teaching Excellence, University of Waterloo

Research Highlights

- Built computational models of human exploration and planning by integrating hierarchical Bayesian modeling, large-scale behavioral data analysis, and approximate planning algorithms (*Plos Comp. Bio* 2024; Annual highlight in *Nature Computational Science 2021*; *NeurIPS LAW Workshop* Spotlight 2025;)
- Developed human-inspired models plan cost-effective compositional policies using program induction and approximate Bayesian inference (*NeurIPS* 2025 Spotlight; US Patent 2025;)

Publications (since 2020)

Articles

S. Dolek, M. Radovanovic, R. Gonzales, J. Sommerville, and **M. Kryven**, "Think outside the box: Making up causal hypotheses from unreliable evidence," *Proceedings of the Annual Meeting of the Cognitive Sciences Society*, vol. 47, 2025.

M. Kryven, W. Cole, A. Curtis, and K. Ellis, "Cognitive maps are generative programs," *Proceedings of the Annual Meeting of the Cognitive Sciences Society*, 2025.

W. T. Piriyakulkij, Y. Liang, H. Tang, A. Weller, **M. Kryven**, and K. Ellis, "Poe-world: Compositional world modeling with products of programmatic experts," *Advances in Neural Information Processing Systems (NeurIPS)*, 2025, **Spotlight**.

J. Qin, A. Yang, Z. Xu, C. Wyeth, and **M. Kryven**, "Planning with generative cognitive maps," *Advances in Neural Information Processing Systems (NeurIPS) LAW Workshop: Bridging Language, Agent, and World Models for Reasoning and Planning*, 2025, **Spotlight**.

X. Zeng, F. Rudzicz, and **M. Kryven**, "Social behavior and strategic adaptation in large language models," *Advances in Neural Information Processing Systems (LAW Workshop: Bridging Language, Agent, and World Models for Reasoning and Planning)*, 2025.

M. Kryven, S. Yu, M. Kleiman-Weiner, T. Ullman, and J. Tenenbaum, "Approximate planning in spatial search," *PLOS Computational Biology*, vol. 20, no. 11, e1012582, 2024.

S. Acquaviva, Y. Pu, **M. Kryven**, T. Sechopoulos, C. Wong, G. Ecanow, M. Nye, M. Tessler, and J. Tenenbaum, "Communicating natural programs to humans and machines," *Advances in Neural Information Processing Systems*, vol. 35, pp. 3731–3743, 2022.

S. Sharma, A. Curtis, **M. Kryven**, J. Tenenbaum, and I. Fiete, "Map induction: Compositional spatial submap learning for efficient exploration in novel environments," *International Conference of Learning Representations*, 2022.

T. Shu, A. Magaro, **M. Kryven**, T. Ullman, and J. Tenenbaum, "Social attribution guides similarity judgment of abstract scenes," *Journal of Vision*, vol. 22, no. 14, pp. 3644–3644, 2022.

C. Bongiorno, Y. Zhou, **M. Kryven**, D. Theurel, A. Rizzo, P. Santi, J. Tenenbaum, and C. Ratti, "Vector-based pedestrian navigation in cities," *Nature Computational Science*, vol. 1, no. 10, pp. 678–685, 2021.

M. Kryven, T. D. Ullman, W. Cowan, and J. B. Tenenbaum, "Plans or outcomes: How do we attribute intelligence to others?" *Cognitive Science*, vol. 45, no. 9, pp. 13–41, 2021.

Y. Qian, **M. Kryven**, T. Gao, H. Joo, and J. Tenenbaum, "Modeling human intention inference in continuous 3d domains by inverse planning and body kinematics," *arXiv preprint arXiv:2112.00903*, 2021.

Y. Pu, K. Ellis, **M. Kryven**, J. Tenenbaum, and A. Solar-Lezama, "Program synthesis with pragmatic communication," *Advances in Neural Information Processing Systems*, vol. 33, pp. 13 249–13 259, 2020.

T. Shu, **M. Kryven**, T. D. Ullman, and J. Tenenbaum, "Adventures in flatland: Perceiving social interactions under physical dynamics.," *Proceedings of the Annual Meeting of the Cognitive Science Society*, 2020.

L. Tian, K. Ellis, **M. Kryven**, and J. Tenenbaum, "Learning abstract structure for drawing by efficient motor program induction," *Advances in Neural Information Processing Systems*, vol. 33, pp. 2686–2697, 2020, **Oral presentation**.

Patents

M. Kryven, "Systems and methods for controlling autonomous robots using machine learning based planning," 10007674-1USPR, Feb. 2025.

Articles Under Review

M. Geva-Sagiv, S. Jun, **M. Kryven**, J. Tenenbaum, E. D. Boorman, R. O'Reilly, J. J. Lin, I. Saez, and C. Ranganath, "Fronto-hippocampal synchronization in rapid spatial learning in humans," 2026.

Y. Guo, D. Hunag, X. Hu, **M. Kryven**, and Y. Pu, "Language-to-function synthesis via embodied affordance validation," (joint senior author), 2026.

P. Zhao, I. Zareski, L. ElAmouri, E. Sansone, B. Zhao, and **M. Kryven**, "Modelling compositional concept leraning as perdictive program synthesis," (joint senior author), 2026.

Refereed Poster Abstracts

R. Gonzales and **M. Kryven**, "Music as programs: Discovery of musical structure via program induction," ISMIR (Workshop paper), 2025.

N. Vlavianos, T. Nagakura, and **M. Kryven**, "Human information seeking in architectural spaces simulated in virtual reality," in *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 44, 2022.

G. Ecanow, C. Wong, S. Acquaviva, Y. Pu, **M. Kryven**, and J. Tenenbaum, "Core knowledge objects in reasoning and language use for highly abstract inductive tasks," in *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 43, 2021.

Z. Yang, **M. Kryven**, H. Shrobe, and J. Tenenbaum, "Modeling human planning in a life-like search-and-rescue mission," in *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 43, 2021.

M. Kryven, S. Croom, B. J. Scholl, and J. Tenenbaum, "Look out, it's going to fall!: Does physical instability capture attention and lead to distraction?" In *Proceedings of the Annual Meeting of the Cognitive Science Society*, 2019, p. 3500.

Selected Invited Talks

2026	Approximate Bayesian inference for problem-solving *University of Waterloo, Universal Algorithmic Intelligence Group*
2025	Where Human Intelligence Meets Artificial Intelligence *Cohere - Industry presentation* Reverse-engineering real-world priors *Dalhaousie University, Department Seminar, Psychology and Neuroscience*
2024	Modeling cognition in real world *University of Toronto, Department of Psychology* Modeling cognition in real world *University of Waterloo, Department of Psychology* Cognitively-inspired AI *Dalhouise University, Department of Computer Science* Cognitively-inspired AI *University of Waterloo, Department of Computer Science* Cognitively-inspired AI *Leiden University, Computer Science Institute*
2021	Modeling human planning under uncertainty, *Harvard, Department of Neuroscience* Modeling human planning under uncertainty, *MIT, Department of Brain and Cognitive Sciences*
2020	Probabilistic Programming, *Lecture at CBMM Summer School, Woods Hole*
2019	Plan-generation and social attribution, *Boston College, Department of Psychology* Plan-generation and social attribution, *Harvard, Department of Neuroscience* Research Methods, *Lecture at CBMM Summer School, Woods Hole*
2018	Planning under uncertainty. *Yale University, Department of Neuroscience* Decision-making experiments, *Demo at MIT Museum* Perception as Inference, *Invited talk at MIT Museum* Perception as inference - what visual illusions tell us about the brain. Guest lecture, *Suffolk University, The New England School of Art and Design* Planning under uncertainty. *MIT, CSAIL*
2017	Value Based Decision-Making under Uncertainty. *MIT, Department of Brain and Cognitive Sciences*
2016	What do people mean by intelligence? *Future of Humanity Institute, Oxford*

Awards

2025	NSERC Discovery Grant $180 000
	NSERC Discovery New Investigator Supplement $12 500
2019	MIT CBMM Research Award $56 000
2018	'Beyond the Ivory Tower' Writing for The Public Workshop, Northwestern University $2100
2017	Doctoral Thesis Award, University of Waterloo, $5000, 3 awarded annually.
2016	Women in Machine Learning Grant $2000
	Cheriton Scholarship, University of Waterloo, 9 awarded annually. $20 000
2015	Distinguished Teaching Assistant Award, University of Waterloo, $500
2014	Graduate Excellence Award in Computer Science, University of Waterloo, $10 000

Active Collaborations & Partnerships

Non-profit	Consultant to Simons Foundation
Academic	I maintain active research collaborations with: Tianmin Shu (Johns Hopkins), Kevin Ellis (Cornell), Yewen Pu (Nanyang Technological University), Bonan Zhao (The University of Edinburgh), Matin Rohrmeier (EPFL), Emanuele Sansone (MIT), Jessica Somerville (UofT)
Industry	Consultant to Site 20/20, National Research Council (NRC), Canada. Advised on data systems design, including data integration and analytics pipelines, for applied industrial projects.

Mentorship

2024-2026	PhD: Cole Wyeth (University of Waterloo), Robie Gonzales, Xijie Zeng (Dalhousie University), MSc: Max Mascini (Dalhousie University), Ziheng Xu (Cornell) Undergraduate: Farzin Ahmadi, Janna Kubursi, Luisa El Amori, Ivan Zareski, Aryaman Pandey, Ryan McArthur (Dalhousie University) Jeffrey Qin, Albert Yang (University of Waterloo), Akshit Sinh (IIT)
2018-2022	Yingdong Quian (MSc, now at Google), Suhyoun Yu (PhD, now at Amazon), Zhutian Yang (MSc, now PhD student at MIT), Nik Vlavianos (PhD, now startup CEO), Sughanda Sharma (PhD, now at Nvidia)

Teaching

2024-2026	Instructor: Computational Cognitive Science (graduate), Machine Learning (3-d year undergraduate), Data Science (3-d year undergraduate)
2019-2022	Teaching Assistant, MIT: Computational Cognitive Science.
2019-2020	Teaching Assistant, Marine Biological Laboratory: Brains, Minds and Machines Summer School
2012-2017	Teaching Assistant, University of Waterloo: Societal Implications of Computing, Introduction to Artificial Intelligence, Real-Time programming, Introduction to Computer Graphics

Service

Conferences / Workshops	Co-Chair CogSci 2027 - 48th Annual Meeting of the Cognitive Science Society Organizer - Computation and Cognition Symposium 2026, Halifax Organizer - CogSci 2026 Workshop on Learning Structured Representations Organizer - CogSci 2026 Contributed Symposium on Music Structure Organizer - NeurIPS 2025 Workshop on Foundation Models for Mind and Brain Organizer - Computation and Cognition Symposium 2025 Halifax Organizing Committee - Atlantic AI Symposium 2025
Grant Leadership:	Nominated Principal Investigator - NFRF-Transformations 2026, submitted Nominated Principal Investigator - NFRF-Explorations 2026, submitted
External PhD Examiner:	2026 Zeng REN, École Polytechnique Fédérale de Lausanne
Journal Reviewer:	Nature Computational Science (3), Nature Scientific Reports (5), Cognitive Neurodynamics (1), Cell (2), Open Mind (2) Neural Computation (1), PNAS (2), Cognition (4), Cognitive Science (6), PLOS Computational Biology (4),
Conference Reviewer:	Neural Information Processing Systems 2017-2025 (5 each year), Cognitive Sciences Society 2016-2025 (5 each year), International Conference on Machine Learning 2024-2026 (5 each year), International Conference on Learning Representations 2025(5), 2024(2) 2023(3), 2021(2), AAAI 2025 (4), International Conference on Robotics and Automation 2021(2)
Outreach	Lecturer, Exhibit designer, MIT Museum (2018-2020)

Personal Information: Canadian and Irish citizen.

January 2026